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Original article

Relative fat mass is a better tool to diagnose high adiposity when compared to body mass index in young male adults: A cross-section study

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SUMMARY

Background and aim: Relative fat mass (RFM) is a new method to estimate whole-body fat percentage in adults using an anthropometric linear equation. We aimed to assess the association between RFM and body fat (BF), evaluated by dual x-ray absorptiometry (DXA) or bioelectrical impedance (BIA), in young male adults.

Methods: Eighty-one young males were assessed for BF fat and free fat mass (by BIA and DXA), waist circumference. BMI and RFM were then calculated from data collected from the subjects. The agreement between BMI and RFM or BIA/DXA was assessed by Pearson's Correlation and Kappa index. Univariate and multivariate linear regression were applied.

Results: Analyzing all the participants together, the correlation between RFM and DXA ($r_{DXA} = 0.90$) or RFM and BIA ($r_{BIA} = 0.88$) were slightly higher than the correlation between BMI and DXA ($r_{DXA} = 0.79$) or BMI and BIA (r_{BIA}: 0.82). When analyzed by BF, low BF (LBF) individuals showed a much higher correlation with RFM ($r_{DXA} = 0.58$; $r_{BIA} = 0.73$) than BMI ($r_{DXA} = 0.24$; r_{BIA}: 0.46). However, subjects with excess BF (EBF) presented similar correlations when comparing RFM ($r_{DXA} = 0.80$; $r_{BIA} = 0.64$) and BMI ($r_{DXA} = 0.78$; $r_{BIA} = 0.64$). In general, RFM presented a higher strength of agreement with DXA and BIA ($k_{DXA} = 0.75$; $k_{BIA} = 0.67$) than BMI ($k_{DXA} = 0.63$; $k_{BIA} = 0.60$). Multivariable linear regression also revealed high associations between RFM and DXA or RFM and BIA ($r^2_{DXA} = 0.85$; $r^2_{BIA} = 0.81$).

Conclusion: Our findings suggest that RFM shows a good correlation and association with BF measured by DXA and BIA in young male adults. Furthermore, RFM seems to be better correlated to BF in LBF individuals when compared to BMI. Therefore, further studies investigating RFM as a tool to assess BF and obesity are motivated.

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1. Introduction

Obesity is a chronic disease characterized by excessive accumulation of adipose tissue, and it is considered one of the most critical risk factors for other chronic non-communicable diseases [1]. In addition to these risk factors, the sustaining obesogenic environment can influence the morbidity and mortality profile [2–4].

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Methods for assessing nutritional status and body composition, such as DXA[1] or BIA,[2] are expensive and not always available to clinicians. On the other hand, the most applied low-cost screening method for obesity is BMI[3] [5]. Although it is widely used toll, there are significant limitations when BMI is applied as a diagnostic tool to the subject level: 1) It does not consider the body fat (BF) localization and proportion [6–8]; 2) Although BMI cut-off values usually used to diagnose obesity have high specificity, they have low sensitivity to identify adiposity as they fail to identify half of the people with excess BF% [9]; 3) Even though a high correlation has been demonstrated between BMI and directly-measured BF, the diagnostic performance is not optimal to identify leanness or excessive BF [10]. That seems to be the case because some individuals have a normal BMI but have increased BF%, characterizing the term known as "normal weight obesity" (NWO), and present an increased risk of developing metabolic disorders [10,11]. Besides, according to these intrinsic limitations to BMI, a more reliable and accurate method is necessary to be used by health professionals. In addition, a method that takes into consideration biological factors as gender, age, and waist circumference (WC), is necessary because these factors are strongly associated with the BF% and distribution [11,12].

The relative fat mass (RFM) was developed as an alternative method to evaluated BF. It is a simple, low cost, and sex-specific anthropometric equation aiming to estimate whole-BF% [13]. The RFM provides results that are easily interpreted and applicable. For instance, whole-BF results are usually more intuitive than the waist-to-height ratio.

The importance of studying body composition has sharply grown in the last few years. Body composition is crucial for the comprehension and decoding of a multitude of pathogenetic processes [14]. BIA and DXA are the two most common body composition assessment techniques extensively used in epidemiological and clinical settings. Also, BIA is considered the most reliable and advanced technology to evaluate body composition in clinical settings, and it has been used in several studies [15–18]. DXA is considered a gold-standard research method for body composition analysis [19–22]. As one of our study focus is the clinical applicability, we choose to include BIA data to improve the applicability of the results. Based on that, we aimed to use RFM in young men and apply it to screen for excess body fat and obesity diagnosis. RFM was then applied to evaluate its correlation and accuracy when compared to BF measured by DXA and BIA analysis. Furthermore, we used the obtained results to diagnose obesity according to BF% assessed by RFM, DXA, or BIA and compared with the results using only BMI. We hypothesized that men with elevated BF mass would be better assessed by RFM when compared to the use of BMI.

2. Materials and methods

2.1. Study sample

This cross-sectional study was conducted in the Federal University of Santa Catarina, the surrounding area of Florianopolis - Santa Catarina - Brazil, and nearby cities from September 2018 to March 2019. Participants were eighty-one healthy male adults (18–35 years old) recruited through flyers posted around the university campus and social media. This study's sample is part of a bigger. Only men were included due to limitations regarding resources to double sample size in case of including women. The eligibility criteria included: 1) BMI 18–34.9 kg/m^2; 2) no smokers; 3) no chronic or metabolic disease diagnosis; and 4) able to lie flat, sit, and stand as required for some tests in this study.

All participants were invited to fill out a standard self-reported questionnaire to evaluate the "health status" and report social-economic variables. The questionnaire collected information about medical history, use of medication, age, smoking, nutritional supplements use, per capita income, and other lifestyle variables. No subjects had a history of diabetes, cardiovascular diseases, dyslipidemias, thyroid diseases, liver and kidney diseases, cancer, or other severe medical illness.

2.2. Data collection

The general guidelines for the participants before anthropometric and body composition exams (DXA and BIA) were: 1) 8–12 h of fasting; 2) to wear a swimsuit; 3) do not perform physical exercise 12 h before the body exam; 3) to avoid caffeine consumption; 4) to be barefoot, and 5) to not wear earrings or any kind of metal (to BIA and DXA analysis). Specifically, to BIA analysis, we kindly asked participants to urinate less than 30 min before the test. All procedures were performed on the same day, early morning (approximately at 8:00 am), and were conducted by well-trained examiners.

Height was measured with a portable stadiometer (Alturexata® - Belo Horizonte, Brazil) to the nearest 0.1 cm. Body mass was measured with a digital scale integrated with the BIA equipment, and by the built-in DXA scan scale, both values are reported when appropriate. Body mass index values were obtained by the ratio between body mass (kg) per height (m) square, resulting in values in kg/m^2 [5]. The WC was measured according to the protocols established by the World Health Organization with the use of an inelastic tape measure (Sanny Medical). The technique for WC calibration was based on the midpoint between the last rib and the iliac crest [23]. The fat mass index (FMI) was calculated by dividing each subject's fat mass (kg) by square of the height (m). Relative fat mass (RFM) was calculated by the equation: 64 - (20 $\times$ (height/WC)) [13]. RFM is dimensionless; however, it is important that both height and WC were in the same unit.

Body composition analysis was performed by Dual-energy X-ray absorptiometry GE® Lunar Prodigy Advance equipment and ENCORE 2004, version 8.10.027 (GE Lunar Corporation, Madison, WI, USA). All body scans were performed by the same trained operator, and the equipment was calibrated at the beginning of each test day. The equipment was also calibrated monthly with the phantom spine scans, according to manufacturer information (Lunar – Lunar Prodigy Advance). The analysis was used as a reference method to measure whole-BF% through the whole-body scan, according to previous studies [24,25]. All subjects completed a total body scan while lying motionless in the supine position. The variables utilized from DXA scans were: 1) Body mass (kg); 2) BF% and BF kg; 3) % android fat mass; 4) % gynoid fat mass; 5) free-fat mass (calculated as bone mineral content plus lean tissue); 6) bone mineral mass (kg). Participants were divided into the lean or excess BF groups according to a previous study in which the DXA body scan cut-off point was set as 22.8% to classify obesity [13].

BIA procedures were performed with InBody® 720 octopolar multifrequency equipment (Biospace, Los Angeles, CA, USA). BIA has eight electrodes, four of which are located on a platform with a coupled scale, two are touched by each foot. The other four electrodes of the equipment were located in two levers, each with two electrodes that are held with the hands. The volunteer was asked to remain in an orthostatic position to perform the exam. The variables utilized from BIA exam were: 1) Body mass (kg); 2) BF% and BF kg; 3) free fat mass (kg); and 4) bone mineral mass (kg). In

[1] Dual-energy X-ray absorptiometry.
[2] Bioelectrical impedance.
[3] Body mass index.

the same way that the subjects were divided by the DXA analysis, the obesity classification was based on the BF% equivalent to 22.8% [13].

The extended version of the International Physical Activity Questionnaire (IPAQ) was applied to determine physical activity [26,27]. The long-form of IPAQ assess detailed information regarding walking, moderate-intensity and vigorous-intensity physical activity in each of the four domains: 1) leisure-time physical activity; 2) domestic and gardening (yard) activities; 3) work-related physical activity; and 4)transport-related physical activity). Using an online tool (webipaq.com.br), three levels of physical activity were proposed to classify populations: a) low; b) moderate, and; c) high. The educational level was defined by years of schooling into three categories: 1) 0–11 years; 2) 12–17 years, and; 3) more than 17 years.

2.3. Ethical approval

Subjects provided written informed consent before all analyses. This study was conducted according to the guidelines laid down in the Declaration of Helsinki, and all procedures involving human subjects were approved by the Human Research Ethics Committee of the Federal University of Santa Catarina (CAAE protocol: 68006617900000121/89244718.6.00000121).

2.4. Statistical analysis

The cross-sectional statistical analyses, including the correlations, strength of agreement, and associations between the exposure (BF analyzed by BIA and DXA) and outcome (RFM) were assessed using the STATA 13.0 (StataCorp LLC, College Station, Texas). Shapiro–Wilk test was applied to assess the distribution of variables. Two groups of analysis were performed: 1) testing all participants as a sole group and 2) separating them into two groups, lean or excess BF. Student's independent t-test was applied to compared BMI, RFM, DXA BF%, BIA BF%, and other variables between lean and excess BF groups. The Mann–Whitney test was used to analyze the fat mass index between low and excess BF individuals according to the BIA division. This data was expressed by the median and interquartile interval. The Chi-square test was applied to categorical variables and expressed by the number of events and percentages.

Pearson's Correlation was used to evaluate the relationship between variables. The strength of the correlation was classified according to the following criteria: very high for *r* of magnitude 1.0 to 0.9, high for 0.9 to 0.7, moderate for 0.7 to 0.5, low for 0.5 to 0.3, and negligible for *r* of magnitude lower than 0.3 [28]. Kappa index was used to evaluate the strength of agreement between variables [29,30]. Values of kappa over 0.81 indicate almost perfect agreement, between 0.61 and 0.80 substantial agreement, 0.41–0.60 moderate agreement, 0.21–0.40 fair agreement, and 0.20 poor agreement [30].

Univariate and multivariable linear regression evaluated the association between DXA/BIA BF% and BMI or RFM on all participants and between lean and excess BF. We used a stepwise regression analysis model to identify variables capable of identifying RFM and BMI values. Independent variables tested included ethnic background, family income, years of schooling, and physical activity levels. Associations resulting in $p < 0.2$ in unadjusted analyses were included in the multiple linear regression model step by step. The adjusted model, according to the description above, was compounded by ethnic background, years of schooling, and physical activity levels. Physical activity level presented an association resulting in $p > 0.2$. However, it was included in the model according to the theoretical model and previous evidence showing that this variable is relevant and associated with BF [31–33]. Wald test was applied to test the maximum likelihood of model variables.

The minimal sample size was based on a *priori* analysis using correlation tests on GPower 3.1 (University of Dusseldorf). For calculation of a two-tail test, alpha value of 0.05, 0.80 of power a conservative 0.5 correlation coefficient was assumed. The calculated minimum sample size to perform the intended analysis corresponded to 46 subjects. We were able to collect data from 81 individuals to perform the analysis listed in this manuscript. The significance level was set at 5% for all tests. Confidence intervals were 95%. Figures were generated on GraphPad Prism 6.0 version (GraphPad Software, San Diego, CA).

3. Results

3.1. Subjects

The characteristics of the subjects involved in this study are presented in Table 1. Data were analyzed using the total sample and divided by DXA or BIA BF% classification for low (LBF) or excess BF (EBF). Subjects participating in the study were from different ethnicities, 54 (66.7%) were self-referred as white, 10 (12.4%) were black, 17 were from other ethnic backgrounds (20.9%). According to DXA BF data, 40 participants were included in the LBF group, and 41 participants were included in the EBF group. However, using BIA BF data, LBF was composed of 58 individuals and EBF by 23 subjects. The mean age was 26.4 ± 4.3 years old. Dividing the sample using DXA BF data for classification, the EBF group had a similar mean age when compared to the LBF group (27.3 ± 4.1 vs 25.5 ± 4.4, $p = 0.06$). Similar values were found using BIA data for classification, according to BF (Table 1).

3.2. Body fat content and other body composition parameters

According to total body scan data using DXA, most of the values for the analyzed variables were higher in the EBF group (Table 1). However, the mean free fat mass in both groups was similar. Total body mass was 87.3 kg ± 15.7 and 70.0 kg ± 8.9 in the EBF and LBF group, respectively. The EBF group presented almost three times more BF mass (kg) than the LBF group (26.4 ± 9.9 vs 9.5 ± 3.4, $p < 0.001$). Therefore, the fat mass index (kg/m^2) was more than two times higher for EBF when compared to LBF (10.0 ± 1.8 vs 4.3 ± 1.4, $p < 0.001$). As expected, RFM value in the was higher in the EBF group when compared to the LBF group (25.2 ± 3.7 vs 17.8 ± 2.8, $p < 0.001$) (Table 1). When the variables were analyzed using BIA data, body mass on the LBF group was 72.5 kg ± 10.7, whereas on the EBF group, it was 94.6 kg ± 13.8 ($p < 0.001$). However, calculated RFM using BIA data were similar when compared to the RFM data using the DXA scan values. Although, there was a significant difference between groups on whole-body free fat mass separating groups by BIA data (LBF = 63.3 ± 8.8 vs EBF = 68.1 ± 7.5, $p < 0.05$).

3.3. Correlation between the body composition assessment methods

When analyzing the whole sample (n = 81), DXA BF % (Fig. 1A) presented a similar correlation with RFM when compared to the BIA vs. RFM (Fig. 1C) ($r_{DXA\ vs\ RFM} = 0.90$ *vs.* $r_{BIA\ vs\ RFM} = 0.88$, respectively) (Table 2). Using DXA scans data to separate groups, LBF had a moderate correlation with RFM ($r_{DXA\ vs\ RFM} = 0.58$), whereas EBF had a high correlation ($r_{DXA\ vs\ RFM} = 0.80$) with RFM. Appling BF % from BIA, we obtained a high correlation between RFM and LBF group and a moderate correlation in the EBF group ($r_{BIA\ vs\ RFM} = 0.73$ and $r_{BIA\ vs\ RFM} = 0.64$, respectively) (Table 2). BIA BF

Table 1
Study subjects physical and biological characteristics.

Variables	DXA			BIA		
	All n = 81	LBF n = 40	EBF n = 41	All n = 81	LBF n = 58	EBF n = 23
Age (years)	26.4 (4.3)	25.5 (4.4)	27.3 (4.1)	26.4 (4.3)	25.8 (4.4)	28.1 (3.8)*
Body mass (kg)[1]	79.0 (15.3)	70.0 (8.9)	87.3 (15.5)*	78.8 (15.3)	72.5 (10.7)	94.6 (13.8)*
Height (m)	1.78 (0.07)	1.78 (0.07)	1.78 (0.07)	1.78 (0.07)	1.78 (0.07)	1.78 (0.06)
BMI (kg/m^2)	24.7 (4.2)	22.0 (2.1)	27.4 (4.2)*	24.7 (4.2)	22.8 (2.4)	29.8 (3.6)*
RFM	21.7 (5.0)	17.8 (2.8)	25.5 (3.7)*	21.7 (5.1)	19.3 (3.4)	27.7 (3.2)*
Whole-body fat (%)	23.0 (10.4)	14.0 (4.5)	31.7 (6.0)*	17.1 (8.6)	12.5 (4.5)	28.6 (4.2)*
Whole-body fat mass (kg)	18.1 (11.3)	9.5 (3.4)	26.4 (9.9)*	14.5 (9.7)	9.2 (3.9)	27.7 (6.9)*
Android fat mass (%)	30.0 (14.2)	17.6 (6.5)	42.1 (7.5)*	–	–	–
Gynoid fat mass (%)	27.8 (9.8)	19.5 (5.6)	35.9 (5.2)*	–	–	–
Android/gynoid ratio	1.0 (0.2)	0.9 (0.2)	1.2 (0.1)*	–	–	–
Fat mass index (kg of fat/m^2)[2]	7.2 (3.2)	4.3 (1.4)	10.0 (1.8)*	7.2 (4.5–9.8)	5.6 (3.9–7.8)	11.5 (10.1–12.2)*
Waist circumference (cm)	85.5 (11.1)	77.4 (4.9)	93.4 (9.7)*	85.5 (11.1)	80.2 (6.6)	98.9 (8.8)*
Whole-body free fat mass (kg)	61.0 (8.1)	61.9 (8.5)	60.2 (7.6)	64.7 (8.7)	63.3 (8.8)	68.1 (7.5)*
Mineral bone mass (kg)	3.2 (0.5)	3.0 (0.5)	3.4 (0.5)*	3.5 (0.5)	3.4 (0.5)	3.8 (0.4)*
Physical activity level[3] (%):						
- Low		7.5 (2.4–21.4)	9.8 (3.6–23.8)		6.9 (2.6–17.3)	13.0 (4.1–34.6)
- Moderate		35.0 (21.6–51.2)	43.9 (29.3–59.6)		43.1 (30.8–56.3)	30.4 (14.8–52.4)
- High		57.5 (41.5–72.1)	46.3 (31.5–61.9)		50.0 (37.1–62.9)	56.5 (35.6–75.4)

LBF: Low body fat; EBF: Excess body fat; DXA: Dual Energy X-ray Absorptiometry; BIA: Bioelectrical impedance; RFM: Relative fat mass; BMI: Body mass index; BF: body fat. Values expressed by the mean and standard deviation. Student's t-test used to compare the difference between lean and excess body fat groups between DXA or BIA body scans. *p < 0.05; 1 Body mass was assessed using integrated DXA scan and BIA scales; 2 Mann–Whitney test expressed by the median and interquartile interval in BIA groups; 3 Qui–Square test: percentages and confidence intervals.

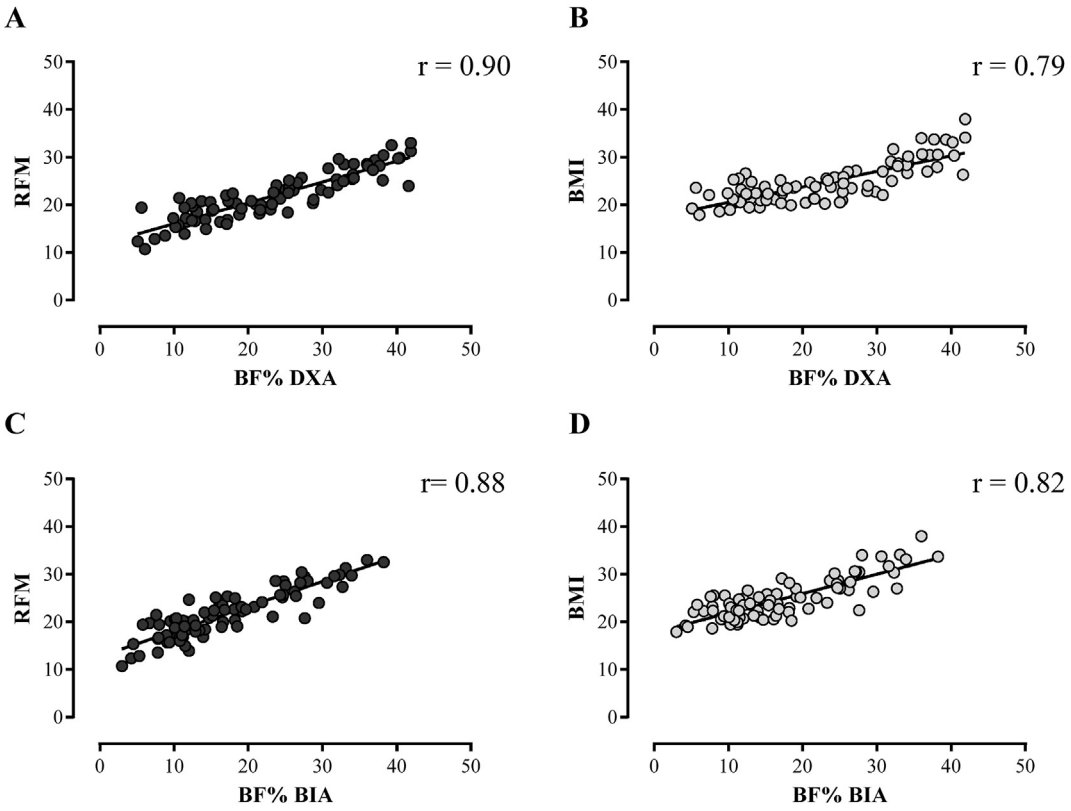

Fig. 1. Pearson's Correlations between body fat percentage (BF%) analyzed by Dual Energy X-ray Absorptiometry (DXA) or BB Bioelectrical impedance (BIA) and Relative Fat Mass (RFM) or Body Mass Index (BMI). A) Correlation between DXA and RFM, r = 0.90; B) Correlation between DXA and BMI, r = 0.79; C) Correlation between BIA and RFM, r = 0.88; and D) Correlation between BIA and BMI, r = 0.82. N = 81.

% showed a higher correlation with BMI than DXA when the whole sample was analyzed (Fig. 1B, D) ($r_{BIA\ vs\ BMI} = 0.82$ *vs.* $r_{DXA\ vs\ RFM} = 0.79$, respectively).

As BF% increased, the correlation coefficient between DXA or BIA and BMI increased. On DXA body scan, the correlation between LBF and BMI is the only relation that returned a negligible and no significant coefficient ($r_{DXA\ vs\ BMI} = 0.24$, $p = 0.131$). However, the DXA BF% in the EBF presented a high correlation with BMI ($r_{DXA\ vs\ BMI} = 0.78$) (Table 2). For BIA BF% data, the correlation between BF% and BMI was low in the LBF individuals ($r_{BIA\ vs\ BMI} = 0.46$), whereas in the EBF subjects, the same correlation was moderate ($r_{BIA\ vs\ BMI} = 0.64$) (Table 2).

Table 2
Correlation between DXA BF% *vs.* RFM or BMI, BIA BF% *vs.* RFM or BIA and RFM *vs.* BMI.

Correlation	DXA			BIA		
	All (n = 81)	LBF (n = 40)	EBF (n = 41)	All (n = 81)	LBF (n = 58)	EBF (n = 23)
BF% vs. RFM	0.90*	0.58*	0.80*	0.88*	0.73*	0.64*
BF% vs. BMI	0.79*	0.24	0.78*	0.82*	0.46*	0.64*
RFM vs. BMI	0.89*	0.68*	0.88*	0.89*	0.71*	0.86*

LBF: Low body fat; EBF: Excess body fat; DXA: Dual Energy X-ray Absorptiometry; BIA: Bioelectrical impedance; RFM: Relative fat mass; BMI: Body mass index; BF%: body fat percentage. *$p < 0.05$.

We also evaluated the correlation between RFM and BMI using total body mass data obtained in the integrated scale for both the DXA scan and BIA. For the whole sample, using DXA scan body mass data, there was a very high correlation between RFM and BMI ($r_{RFM\ vs\ BIA} = 0.89$). When splitting groups by BF, the correlation between RFM and BMI was moderate in the LBF ($r_{RFM\ vs\ BIA} = 0.68$), and very high between in the EBF group ($r_{RFM\ vs\ BIA} = 0.88$) (Table 2). When using BIA scale body mass data, the correlation between RFM and BMI for the whole sample was very high ($r_{RFM\ vs\ BMI} = 0.89$). The separation in groups by BF returned a high correlation for the LBF group ($r_{RFM\ vs\ BIA} = 0.71$), and a very high correlation for and EBF group ($r = 0.86$) (Table 2).

3.4. Strength of agreement between the body composition assessment methods

We evaluated the strength of agreement between the variables through the kappa index (Table 3). In general, when testing for the whole sample, RFM agreed to 87.7% with DXA body scan exam BF%. This represents a substantial agreement. For lean individuals (LBF), RFM agreement corresponded to 100% with DXA BF%. In cases of EBF, the agreement was 82.8%. Testing BIA BF% and RFM, returned an 85.2% agreement. For the LBF group, BIA and RFM showed 82.8% of agreement and the EBF group 91.3% agreement.

Nevertheless, DXA and BMI agreed 81.5% for adiposity classification, showing a substantial agreement. When analyzed by adiposity levels, lean individuals (LBF) presented a 90% agreement between DXA and BMI. However, for individuals classified as excess adiposity (EBF), the agreement was quite lower, 73.2% between DXA and BMI. Using adiposity from BIA, the agreement with BMI, in general, was 81.5%. In the LBF group and EBF, divided by BIA adiposity data, agreements were 75.6% and 91.3%, respectively (Table 3).

3.5. Univariate and multivariable linear regression models of characteristics in different methods, RFM, and BMI data

A strong association was found between RFM and DXA BF% ($r^2_{DXA\ vs\ RFM} = 0.81$; $\beta_{DXA\ vs\ RFM} = 0.44$) (Fig. 2A). However, when the analysis was performed by levels of BF, LBF and EBF, results showed low and moderate associations (LBF: $r^2_{DXA\ vs\ RFM} = 0.34$, $\beta_{DXA\ vs\ RFM} = 0.36$, EBF: $r^2_{DXA\ vs\ RFM} = 0.64$, $\beta_{DXA\ vs\ RFM} = 0.48$) (Table 4). The association between BIA BF% and RFM for the whole sample data was closer to the numbers obtained with DXA data ($r^2_{BIA\ vs\ RFM} = 0.78$, $\beta_{BIA\ vs\ RFM} = 0.52$) (Fig. 2C). For the two different adiposity groups, associations between BIA BF% and RFM were also moderate (LBF: $r^2_{BIA\ vs\ RFM} = 0.53$, $\beta_{BIA\ vs\ RFM} = 0.56$, and EBF: $r^2_{BIA\ vs\ RFM} = 0.41$, $\beta_{BIA\ vs\ RFM} = 0.48$, respectively).

The associations between DXA and BMI or BIA and BMI, when assessed for the whole sample (Fig. 2B, D), where moderate ($r^2_{DXA\ vs\ BMI} = 0.63$; $r^2_{BIA\ vs\ BMI} = 0.66$). When the analysis was stratified by the BF ranges, determination coefficients were directly proportional to the increase in BF% (Table 4). Associations in the LBF group between DXA and BMI or BIA and BMI were low ($r^2_{DXA\ vs\ BMI} = 0.06$, $\beta_{DXA\ vs\ BMI} = 0.11$; $r^2_{BIA\ vs\ BMI} = 0.20$, $\beta_{BIA\ vs\ BMI} = 0.25$, respectively). On EBF, DXA and BMI presented a moderate association ($r^2_{DXA\ vs\ BMI} = 0.61$, $\beta_{DXA\ vs\ BMI} = 0.54$) and a low association between BIA and BMI ($r^2_{BIA\ vs\ BMI} = 0.41$, $\beta_{BIA\ vs\ BMI} = 0.56$) (Supplementary Figs. 1 and 2).

The linear regression model adjusted by ethnic background, education (years), and physical activity level returned a high association with RFM for both DXA and BIA ($r^2_{DXA\ vs\ RFM}$ 0.85; $\beta_{DXA\ vs\ RFM} = 0.43$ and $r^2_{BIA\ vs\ RFM}$ 0.80; $\beta_{BIA\ vs\ RFM} = 0.51$, respectively) (Table 4). Both values found using RFM were higher than those obtained when testing associations using BMI ($r^2_{DXA\ vs\ BMI} = 0.70$, $\beta_{DXA\ vs\ BMI} = 0.35$; $r^2_{BIA\ vs\ BMI} = 0.69$, $\beta_{BIA\ vs\ BMI} = 0.40$). The adjusted model showed that association values between DXA and RFM relatively high for EBF individuals ($r^2_{DXA\ vs\ RFM}$ 0.72; $\beta_{DXA\ vs\ RFM} = 0.53$).

4. Discussion

In the present study, we found a high correlation, substantial agreement, and high association between RFM and DXA or BIA. Furthermore, RFM better assessed excess body fat in individuals not considered obese according to BMI. To our knowledge, this is the first study to make this analysis and this investigation in male young Brazilian individuals.

In our sample, some participants were eutrophic according to BMI, although they presented high adiposity according to the DXA scan. Such finding is relevant for clinical practice and shows that using BMI alone may result in misleading information and diagnosis. As excess BF% has been associated with metabolic dysregulation regardless of body mass, BMI should not be considered as the

Table 3
The strength of agreement between DXA/BIA and RFM/BMI.

	BF %	Kappa index***	Agreement all	Agreement LBF	Agreement EBF
RFM	DXA	0.75	87.7% *(n = 81)*	100% *(n = 40)*	75.6% *(n = 41)*
	BIA	0.67	85.2% *(n = 81)*	82.8% *(n = 58)*	91.3% *(n = 23)*
BMI	DXA	0.63	81.5% *(n = 81)*	90.0% *(n = 40)*	73.2% *(n = 41)*
	BIA	0.60	81.5% *(n = 81)*	75.6% *(n = 58)*	91.3% *(n = 23)*

LBF: Low body fat; EBF: Excess body fat; DXA: Dual Energy X-ray Absorptiometry; BIA: Bioelectrical impedance; RFM: Relative fat mass; BMI: Body mass index; BF: body fat. *** all p-value <0.001. Values of kappa: $\kappa > 0.81$: almost perfect agreement; $0.80 \geq \kappa \leq 0.61$: substantial agreement; $0.61 \geq \kappa \leq 0.41$: moderate agreement; $0.40 \geq \kappa \leq 0.21$: fair agreement; $\kappa \leq 0.20$: poor agreement.

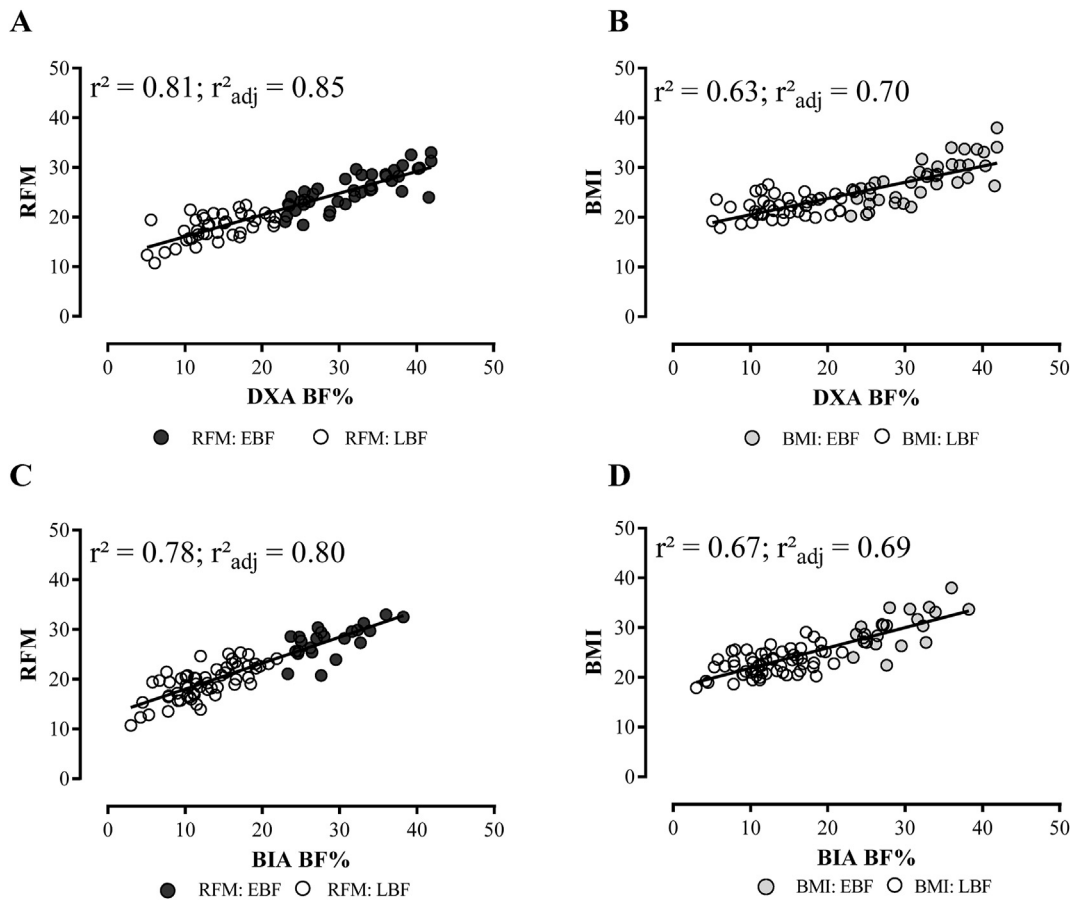

Fig. 2. Simple linear regression between Relative Fat Mass (RFM) or Body Mass Index (BMI) with Dual Energy X-ray Absorptiometry (DXA) or Bioelectrical impedance (BIA). A) Association between DXA and RFM on all sample was very high ($r^2 = 0.81$); B) Association between DXA and BMI had a lower coefficient than DXA *vs.* RFM, with a moderate association ($r^2 = 0.63$); C) BIA, RFM had a high association with the equipment ($r^2 = 0.78$); D) As with DXA, BMI was moderately associated with BIA ($r^2 = 0.66$).

Table 4
Univariate and multivariate linear regression models of characteristics in different methods, RFM, and BMI data.

		Crude model				Adjusted model*			
		β (95%CI)	Standard error	r^2	*p*	β (95%CI)	Standard error	r^2	*p*
RFM	***DXA:***								
	All (n = 81)	0.44 (0.38–0.49)	2.24	0.81	<0.001	0.45 (0.40–0.50)	2.09	0.85	<0.001
	LBF (n = 40)	0.36 (0.19–0.53)	2.28	0.34	<0.001	0.41 (0.26–0.56)	1.95	0.58	<0.001
	EBF (n = 41)	0.48 (0.37–0.60)	2.22	0.64	<0.001	0.53 (0.39–0.67)	2.11	0.73	<0.001
	BIA:								
	All (n = 81)	0.52 (0.46–0.58)	2.38	0.78	<0.001	0.51 (0.45–0.58)	2.36	0.80	<0.001
	LBF (n = 58)	0.56 (0.42–0.70)	2.38	0.53	<0.001	0.58 (0.44–0.73)	2.30	0.61	<0.001
	EBF (n = 23)	0.48 (0.22–0.74)	2.47	0.41	0.001	0.54 (0.23–0.85)	2.59	0.54	0.002
BMI	***DXA:***								
	All (n = 81)	0.33 (0.27–0.38)	2.60	0.63	<0.001	0.35 (0.29–0.40)	2.44	0.70	<0.001
	LBF (n = 40)	0.11 (−0.03–0.26)	2.04	0.06	0.131	0.17 (0.03–0.31)	1.78	0.38	0.02
	EBF (n = 41)	0.54 (0.40–0.67)	2.64	0.61	<0.001	0.52 (0.40–0.74)	2.57	0.69	<0.001
	BIA:								
	All (n = 81)	0.41 (0.34–0.47)	2.46	0.67	<0.001	0.40 (0.33–0.47)	2.49	0.69	<0.001
	LBF (n = 58)	0.25 (0.12–0.38)	2.19	0.21	<0.001	0.30 (0.16–0.43)	2.16	0.31	<0.001
	EBF (n = 23)	0.56 (0.25–0.86)	2.85	0.41	0.001	0.58 (0.20–0.96)	3.2	0.47	0.005

LBF: Low body fat; EBF: Excess body fat; RFM: Relative fat mass; BMI: Body mass index; DXA: Dual Energy X-ray Absorptiometry; BIA: Bioelectrical impedance. r^2: Determination coefficient. * Adjusted model to ethnic background, physical activity and instruction level.

only parameter to diagnose obesity for patient care settings [9]. In contrast, we found that RFM is more accurate than BMI. The BMI values for the diagnose of obesity have high specificity but low sensitivity to identify adiposity, as they fail to identify with accuracy people with excess BF [9]. Nevertheless, RFM seems to be is a low-cost alternative with better diagnostic accuracy for obesity, as shown by others [13,34] and now by us.

The NWO is a concept that has emerged to question the relationship between obesity and metabolism. NWO is characterized by a body mass, and BMI considered in the normal range, but

increased BF% [11]. Studies have shown that individuals with normal body mass can have elevated BF%. Consequently, they have a higher risk of developing metabolic disorders [10,35,36]. In our study, we found that the association between RFM and BIA/DXA presented higher determination coefficients than those obtained by the association against BMI. This means that some "eutrophic" individuals in our study, according to BMI, presented a high BF%, indicating cases of NWO. Also, RFM association with BIA/DXA presented higher coefficients for both the low and high adiposity groups. Therefore, it seems reasonable to assume that RFM is a better tool to assess BF without having to rely on an expensive piece of equipment.

The use of BMI for classification of adiposity status and disease risk is based on epidemiological associations of BMI with morbidity and mortality [37]. Although easily applicable in clinical practice, BMI can misrepresent the adiposity [38]. Kennedy et al. found a large discrepancy between BMI and adiposity defined by DXA [37]. Other anthropometric markers, such as waist-to-hip ratio, waist-to-height ratio, are also widely used to assess adiposity-related outcomes and metabolic complications [39,40]. However, they were not the focus of our study.

It is known that BMI classification of body fat excess and obesity is defined by statistical criteria, disregarding biological characteristics, such as age and sex. However, RFM can account for biological sex and WC on the equation [13]. Including WC to this analysis adds the metabolic risk component associated with the accumulation of fat in the abdominal region. Based only on BMI, an NWO individual would be paradoxically classified as a low risk of developing chronic diseases [12,41,42]. In this regard, a recent populational study demonstrated that high RFM provides high predictability for dyslipidemias and metabolic syndrome when compared to BMI [34]. As described in the study, RFM emerges as an interesting indicator to screen for outcomes related to chronic non-communicable diseases [34]. The implementation in the clinical practice of anthropometric indicators, in addition to BMI, is important when the physiological and metabolic impact of BF in the abdominal region is taken into consideration. The increase in visceral adipose tissue is associated with several metabolic disturbances [43–45].

Woolcott and Bergman argue that, compared with BMI, RFM can better predict whole-BF%, measured by DXA. RFM showed better accuracy than the BMI and had fewer false-negative cases of obesity defined by BF %. More importantly, RFM reduced total obesity misclassification [13]. The RFM performance to estimate DXA-measured BF% was overall more consistent than BMI among women and men, across ethnic groups, young, middle-aged, and older adults. Although in the study, the accuracy of RFM was lower among individuals with lower BF, something that we also found in our study, the accuracy of the RFM increases with the percentage of fat measured by the DXA [13]. Noteworthily, there are no established parameters for obesity diagnosis based on BF% specific for the Latin American population. That is an issue limiting further extrapolations of our findings.

Our study has some limitations. DXA estimative of BF and BIA measurements are both sensitive to body hydration [46]. However, we made our best to control such variables making sure our subjects followed all study instructions with accuracy. Another point raised by some authors has to do with the potential limitations of using DXA to estimate BF. On this subject, LaForgia and collaborators [19] indicate that DXA may provide reasonable cross-description data for obese cohorts. Also, DXA has been employed as the "gold standard" research method for body composition analysis, and it seems to be a reliable method to estimate total and regional BF featuring improved feasibility, high accuracy, and reproducibility [20,21,47]. One major limitation of our study is that cut-off points were based on the North American population because there are no cut-off points established for the Brazilian population. However, other studies using DXA and BIA applied these same cut-off points before [22,48,49]. Therefore, future studies generating cut-off points adapted to the Brazilian population are necessary to perfectly address this issue. In this way, the classification could be more accurate. Furthermore, someone might argue that RFM classification is also based initially on a North American population study [13]. Nevertheless, the present study aims to test if RFM results agree with gold standard measurements for a sample extracted from a Brazilian population. This study had a relatively small sample size, and it was composed of males, aged between 18 and 35 years and BMI range corresponded to 18.5–34.9 km/m^2. However, all statistical power calculations performed by us resulted in values higher than 95% for the relevant outcomes regarding this sample. Our study, as a preliminary study with a sample of the Brazilian population, was carried out in men only due to limitation of resources available to double sample size, which would be necessary to perform a proper analysis in women. Therefore, based on our results, we encourage future research that aims to evaluate other populations, certainly including women and men with different biological characteristics (i.e., different BMI and age ranges). Finally, readers should be aware of the assumptions and intrinsic limitations regarding mathematical models testing correlations, associations and other statistical techniques applied in this study, as those can also limit the applicability of our findings.

Our study had some strengths. We used two different body composition methods, DXA and BIA. We also tested a sample with a wide range of BF%, not limiting the evaluation to one biotype. The assessment and adjustment for relevant confounding factors as the level of physical activity (PA) is also a strength. PA was evaluated by IPAQ, a widely and validated tool to measure the PA level [50,51]. PA adjustment is essential because there is an inverse association between PA and BF [31–33].

In conclusion, our study shows that RFM better performs to screen for high adiposity normal-weight individuals with BMI <30 kg/m^2. Such finding is relevant since people with normal-weight obesity might be at risk of developing metabolic diseases in the future. Also, RFM results were consistent with measuring body composition methods that some professionals might not have access (e.g., DXA and tetrapolar BIA). In addition, RFM is a simple, low-cost, and non-invasive method that contributes to the diagnosis of excess BF. We suggest the use of RFM as a complementary method of assessing body composition, both in clinical practice and trials. We also encourage further studies that apply the RFM as an anthropometric indicator to evaluate obesity and body fat content.

Author contributions

Cinthia R Corrêa: Conceptualization and design of the study, Acquisition of data, Methodology, Formal analysis and interpretation of data, Investigation, Writing – Original Draft.

Natália P. S. Formolo: Conceptualization and design of the study, Acquisition of data, Investigation.

Talissa Dezanetti: Investigation, Methodology, Acquisition of data.

Guilherme F. F. Speretta: Supervision, Writing - Review & Editing, Revising it critically for important intellectual content.

Everson A. Nunes: Conceptualization and design of the study, Supervision, Project administration, Writing - Review & Editing, Funding acquisition, Revising it critically for important intellectual content, Final approval of the version to be submitted.

Declaration of competing interest

There are no financial, professional, or personal conflicts of interest for any of the authors.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2020.12.009.

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